

8	(a)	(i)	Radioactive decay is the <u>spontaneous emission of particles or radiation</u> [B1] <u>from an unstable nucleus so that it becomes more stable</u> [B1].
		(ii)	Random decay means it is <u>impossible to predict which nucleus will decay next</u> [B1]. There is a <u>constant probability that a nucleus will decay in any fixed period of time</u> [B1]
		(iii)	Spontaneous decay means the decay process is <u>not affected by external conditions</u> . [B1]
	(b)	(i)	α : mass $\approx 4u$, charge = $+2e$ [B1] β : mass = $9.11 \times 10^{-31} \text{ kg}$, charge = $-e$ [B1] γ : zero mass, no charge [B1]
		(ii)	Alpha and beta radiation have opposite charges, so will experience <u>magnetic forces in opposite directions based on Fleming's Left Hand Rule when moving into a uniform magnetic field in the same direction</u>, and get <u>deflected in opposite directions</u>. [B1] Gamma radiation has no charge so <u>will not experience any magnetic force</u> and pass through a uniform magnetic field <u>undeflected</u>. [B1]
	(c)	(i)	Beta decay [B1]
		(ii)	Energy released $= (\text{mass of } ^{14}\text{C nucleus} - \text{mass of } ^{14}\text{N nucleus} - \text{mass of beta particle}) c^2$ $= (13.999950 \times 1.66 \times 10^{-27} - 13.999234 \times 1.66 \times 10^{-27} - 9.11 \times 10^{-31})(3.00 \times 10^8)^2 \text{ [C1]}$ $= 2.49804 \times 10^{-14} \text{ J}$ $= 0.156 \text{ MeV [A1]}$
	(d)	(i)	The <u>average time taken for the number of carbon-14 nuclei in a sample to decrease by half of the original number is 5730 years</u> . [B1]
		(ii)	Decay constant = $\frac{\ln 2}{\text{half life}} = \frac{\ln 2}{5730} = 1.209 \times 10^{-4} \text{ year}^{-1}$ [B1 for answer, B1 for appropriate unit]
		(iii)	For the sample, $^{14}\text{C} : ^{12}\text{C} = \frac{6.0 \times 10^5}{(1.0 \times 10^{20} - 6.0 \times 10^5)} = 6.00 \times 10^{-15} \text{ [C1]}$ Let t be the age of the bone fragment $\frac{6.00 \times 10^{-15}}{1.3 \times 10^{-12}} = \left(\frac{1}{2}\right)^{\frac{t}{5730}} \text{ [C1]}$ $t = 44000 \text{ years [A1]}$

	(e)		After 35000 years, the activity of the object will be too low and <u>the count rate due to it will be indistinguishable from the background count rate</u> [B1]
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